

CONVERSATIONS FOR LEARNING: DESIGNING PERSONIFIED HISTORICAL CHATBOTS TO ENHANCE CRITICAL THINKING IN K-12 STUDENTS

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ABSTRACT

This study explores the design and implementation of personified historical chatbots for K–12 history education. Traditional history classrooms often emphasize rote memorization and factual recall, limiting opportunities for inquiry, reflection, and student engagement. To address these challenges, we developed persona-based chatbots modeled after historical figures such as Mahatma Gandhi, Albert Einstein, Benjamin Franklin, Leonardo da Vinci, and Chanakya. The system integrates large language models (LLMs) with retrieval-augmented generation (RAG) to provide contextually accurate and persona-aligned responses. Guided by dialogic pedagogy, the chatbots encourage students to ask questions, reflect critically, and engage in meaningful dialogue with historical personas. The findings from the case study of a student interaction with a personified chatbot revealed that the dialogic interactions progressed from factual inquiries to reflective, ethical, and hypothetical questions, highlighting the potential of persona-based chatbots to support curiosity, perspective-taking, and the development of critical thinking skills in history learning.

KEYWORDS

Personified Chatbots, History Education, K–12 Learning, AI in Education, GenAI

1. INTRODUCTION

History is a core subject in schools, enabling students to understand the individuals, communities, and societies that shaped the past. Learning history goes beyond memorizing facts and milestones; it involves analyzing events through a critical, historical lens while fostering future-oriented thinking. Research highlights that history education develops historical consciousness, civic responsibility, and transferable skills such as sourcing, contextualization, and corroboration (Wineburg, 2018; Nuttall, 2021; Cowgill & Waring, 2017).

Despite its significance, history teaching faces persistent challenges. Traditional approaches, including textbook-driven instruction, lecture-based delivery, and assessments focused on factual recall, often limit inquiry and reflection (Purmintasari et al., 2024; Callaway, 2020). These practices reduce opportunities for students to critically engage with historical perspectives, as history is frequently presented as fixed knowledge (Rivero et al., 2022). Teachers also encounter constraints such as large class sizes, standardized curricula, and assessment pressures, which restrict interactive and personalized learning (Rashadi & Ahmad, 2019). Consequently, history learning often becomes confined to memorization and pre-packaged narratives rather than inquiry and critical engagement.

To address these challenges, active learning strategies have been promoted to engage students in inquiry, collaboration, and problem-solving. Approaches such as game-based learning, role-play, and dialogic pedagogy have demonstrated improvements in participation, critical thinking, perspective-taking, and deeper historical understanding (Moseikina, 2022; Stack, 2023; Shargel & Katz, 2025; Prendergast, 2024). Among these, dialogic pedagogy emphasizes classroom dialogue to promote reasoning, reflection, and historical inquiry. In Taiwanese history classrooms, teachers combine monologue and dialogue through short- and long-term discussions to foster critical engagement and historical reasoning. Such approaches enhance student participation, support democratic values, and develop skills in argumentation and evaluation of truth (Alexander, 2018). However, active learning strategies, including dialogic pedagogy, face limitations.

Challenges include scaling, variability in teacher practices, classroom management, technological integration, and rigid curricula (Akçayır & Akçayır, 2018; Shulman, 2018). These limitations highlight the need for careful design, ongoing support, and balanced integration to ensure the effectiveness of active learning in history classrooms.

Recent research highlights that chatbots built on archival records can enable meaningful conversational interactions with historical figures. By drawing directly from primary and secondary sources, these systems allow learners to explore historical perspectives by asking questions in museum and educational contexts (Qian et al., 2022). The “Social bots of conviction” have been shown to promote historical empathy and perspective-taking by facilitating reflective group discussions about past events and their connections to contemporary contexts (Petousi et al., 2021). Similarly, ChatGPT simulations of historical figures, such as Simón Bolívar, have demonstrated improvements in students’ critical thinking, historical understanding, and analytical abilities through dialogic interaction (Santamaria-Velasco et al., 2025). Although effective, some of these systems are limited by providing only archival text responses rather than generative dialogue, facing challenges related to information bias (Santamaria-Velasco et al., 2025). By enabling learners to interact with historical narratives in a dialogic and participatory way, personified chatbots can transform history learning from a passive exercise in memorization into an active process of critical reflection and perspective-taking.

Building on this potential, the present study explores the design of persona-based historical chatbots using Retrieval-Augmented Generation (RAG) and large language models to provide personalized, dialogic interactions while presenting information from well-integrated resources. The study also investigates how students engage with these systems and aims to contribute to the broader goal of creating effective, interactive history learning experiences that foster critical thinking

2. LITERATURE REVIEW

2.1 Teaching Learning Methodology in History

History classrooms have traditionally relied on lecture-based, textbook-centered instruction, where teachers transmit content and students are expected to recall factual knowledge. Relying on third-person narration, history is often presented as fixed and complete, limiting opportunities for students to question sources, perspectives, or interpretations (López-Fernández et al., 2023). Consequently, learners often engage passively, focusing on memorization rather than developing historical thinking skills such as contextualization, sourcing, corroboration, and perspective-taking (Rivero et al., 2022; Perrotta & Bohan, 2013). Assessment practices emphasizing factual recall further reinforce this passivity, discouraging critical analysis and independent interpretation (Purmintasari et al., 2024; Norakma Mohd Daud et al., 2015; Rashadi & Ahmad, 2019). Traditional teaching methods, constrained by standardized curricula, time limitations, and exam pressures, often fail to cultivate higher-order thinking, creativity, or meaningful engagement with historical content.

To address persistent challenges of passivity and rote learning in history classrooms, active learning strategies have been increasingly promoted as alternatives that foreground inquiry, collaboration, and reflective reasoning. Dialogic pedagogy, for instance, shifts classroom talk from recitation to purposeful questioning, argumentation, and collective meaning-making. By creating structured opportunities for students to justify claims, challenge peers, and build on ideas, dialogic teaching has been shown to improve both engagement and achievement (Alexander, 2018; Howe et al., 2019; Prendergast, 2024). In East Asian contexts, dialogic approaches further support historical thinking by encouraging students to interrogate sources and explore multiple perspectives while respecting cultural communication norms (Chang, Wegerif, & Hennessy, 2023; Geographical Association, 2025). Alongside classroom dialogue, game-based learning (GBL) has emerged as a powerful tool for making historical processes more tangible. Strategy games such as *Napoleon: Total War* allow learners to grapple with causation, contingency, and scale by placing them in the position of historical decision-makers (Stack, M., & Bunt, B., 2023). Students report high enjoyment and a sense of agency in such environments, though effective learning requires structured pre-briefs and reflective debriefs to connect gameplay with disciplinary reasoning (Hellerstedt A. 2025).

Research shows that role-play strategies complement digital games and can enhance empathy, perspective-taking, and moral reasoning when supported by evidence-based preparation and post-activity reflection (Khan. (2022).). Also, both in-person and virtual, have been shown to improve critical thinking, perspective-taking, and higher-level understanding of complex events (Shargel & Katz, 2025).

Collectively, the above-mentioned strategies shift history teaching from teacher-centered transmission toward student-centered inquiry. They enable learners to move beyond surface-level recall and instead engage in sourcing, corroboration, and contextualization while fostering critical thinking and analytical reasoning. Despite their benefits, active learning strategies face limitations, including high preparation demands, classroom management challenges, and technological requirements, including large class sizes, rigid curricula, assessment pressures. Similarly, sustaining dialogic interaction demands skilled facilitation and cultural sensitivity.

Technology-enhanced solutions like chatbots can help address these challenges by providing personalized feedback, adaptive scaffolding, and support for self-regulated learning, thereby increasing engagement and motivation (Chang et al., 2023; Baskara, 2023). Additionally, LLM-based chatbots using Socratic questioning can foster critical thinking, reflection, and intellectual autonomy, while support introverted students and maintain privacy and accessibility (Hobert & Meyer von Wolff, 2019; Lucile, 2025).

2.2 Chatbots in Education

A chatbot is a computer program or system that engages users in human-like conversation through text or voice interfaces, often powered by artificial intelligence and natural language processing (Shawar & Atwell, 2007; Mauldin, 1994). Their primary value lies in providing real-time support, enhancing engagement, and offering personalized learning experiences (Clarizia et al., 2018). Kuhail et al. (2023) classify educational chatbots into four roles: teaching agents, peer agents, teachable agents, and motivational agents, each supporting learning in different ways.

Kuhail et al. (2022) emphasized that chatbots can act as intelligent assistants, offering real-time responses to students' queries and encouraging active engagement with the content. Similarly, Labadze, L. et al. (2023), noted that chatbots are capable of adapting their interactions to align with individual learning preferences, thereby catering to diverse student needs. These features not only enhance access to information but also support the development of critical thinking and inquiry-driven learning.

In history education, some research studies have shown how chatbots can serve as powerful tools to enhance learning. Lee Bih Ni (2024) highlights that chatbots provide real-time, personalized responses and create immersive experiences that help students deepen their historical understanding and improve retention, particularly in K–12 contexts. Dimitra Petousi (2021) designed a “social” Bot of Conviction (BoC) for high school history education, aimed at sparking group reflection rather than just delivering information. The findings showed that the BoC effectively promoted historical empathy by encouraging perspective-taking, constructive dialogue, and linking past events to present issues.

However, many chatbots rely on scripted responses, leading to shallow dialogue and limited adaptability (Galitsky et al., 2019), while their lack of emotional intelligence reduces meaningful interaction (Wang et al., 2023). Recent developments in generative AI have begun to address these issues where language models can conduct more fluid, context-aware conversations, increasing personalization, engagement, and learner autonomy. Nevertheless, these systems introduce new challenges, including potential misinformation, ethical transparency, and student over-reliance. Hybrid models, which leverage Retrieval-Augmented Generation (RAG) to incorporate information from provided documents alongside LLM capabilities, are being explored to mitigate these risks.

3. SOLUTION APPROACH

3.1 Personified Chatbot

The proposed solution for the particular research problem is a personified educational chatbot developed by an Educational Research Lab in India, designed to embody prominent historical and polyglot personalities such as Mahatma Gandhi, Albert Einstein, Benjamin Franklin, Leonardo da Vinci, and Chanakya. The system is a web-based application that goes beyond conventional chatbots by offering an immersive dialogic experience rather than only factual responses. The chatbots are modeled as polyglot personas to represent their multifaceted knowledge and diverse expertise, allowing students to engage with each historical figure in a manner that reflects their unique worldview and interdisciplinary thinking.

3.2 Personification of Chatbot

A central feature of the design is the persona-based interaction model. In this system, a “persona” refers to a digital entity endowed with a recognizable identity, modeled after historical, fictional figures. For this prototype, the focus is on polyglot personas that symbolize linguistic diversity and intercultural competence. The persona layer defines the chatbot’s tone of communication, response style, and cultural framing, ensuring that interactions feel authentic and engaging. This is implemented through prompting to the LLM, allowing the system to generate responses aligned with each persona’s characteristics. By embedding personas into the chatbot, the design leverages the human tendency to anthropomorphize technology. From a human-computer interaction perspective, prior research demonstrates that systems exhibiting social presence and relatable characteristics foster stronger learner engagement, motivation, and trust (Smestad, 2019). A snapshot of the persona-based chatbot interface is presented in Figure 1 to illustrate the persona-based interaction model.

3.3 Design of the Personified Chatbot

The design architecture of the chatbot integrates Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) to balance conversational flexibility with factual accuracy (Fan, W. 2024; Yang, D., 2023). While the LLM ensures natural, context-sensitive dialogue, RAG grounds responses in curated, authoritative sources. This dual-layered design addresses the challenge of hallucinations in generative models, thereby enhancing trustworthiness and positioning the system as a reliable educational companion.

The architecture illustrated in figure 2 represents a Chatbot system design built using the large language model (LLM) and Retrieval-Augmented Generation (RAG). This particular chatbot system has two main parts: (1) retriever and (2) generator. The retriever looks for content that matches the user’s query from the existing provided material, whereas the generator uses a pre-trained model (in our case, ChatGPT 3.0), and produces natural and meaningful responses. Translating to this study context, as a student interacts with the chatbot, the process starts with typing a question. For example, a student asked Gandhiji bot, “*why did you choose to fight for freedom*”. This input is first handled by the retriever, which searches through a provided collection of documents. These documents are divided into smaller parts, called chunks, and each chunk is converted into a numerical representation called an embedding. Embeddings capture the meaning of the text in a form the system can compare and store in a vector database. The student’s question is also converted into an embedding, and the system compares it with all stored embeddings to find the most relevant information (Church, K 2024) For this question, the retriever might select chunks related to Gandhi’s experiences in South Africa, his belief in justice, and his commitment to nonviolence.

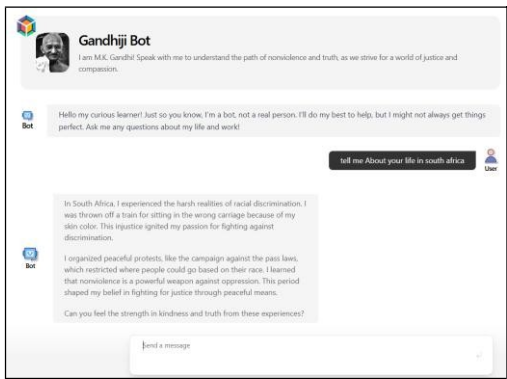


Figure 1. Interface of Persona-based Chatbot representing Gandhiji

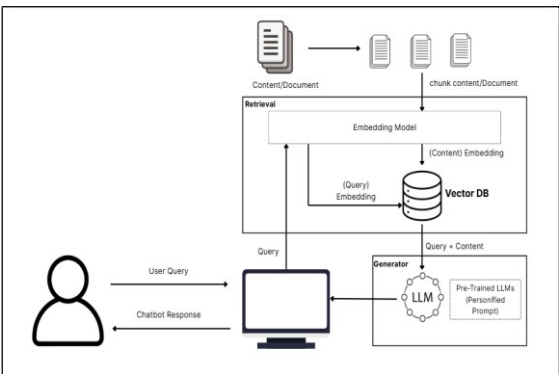


Figure 2. System Architecture of Persona-based Chatbot

Next, the generator, powered by a pre-trained large language model (LLM) such as ChatGPT, combines the retrieved information with the student’s question to produce a coherent and meaningful response. In this case, the chatbot replied: “*Ah, my young friend! The fight for freedom is a journey that lies in the heart and spirit of every human being. I chose to fight for India’s freedom because I....*”.

This sample response demonstrates that the chatbot not only understands the question but also provides a detailed, historically accurate, and contextually rich answer. A key feature of this design is personification, allowing the chatbot to respond as a historical figure, making interactions more relatable and immersive. By merging retrieval-based accuracy with LLM-based fluency, the system supports dialogue-based exploration of historical content while maintaining reliability.

The design of the chatbot is informed by dialogic pedagogy (Alexander, 2018), which frames learning as a process achieved through meaningful conversation. Students' questions and ideas are explored interactively, allowing them to critically engage, ask follow-up "why" or "how" questions, and receive evidence-based responses. For example, when a student asks the Gandhiji chatbot, "Why did you choose to fight for freedom?" The system replies with a historically grounded, contextually rich answer, fostering joint interpretation of historical events. The designed chatbot supports six dialogic principles: collective, reciprocal, supportive, deliberative, cumulative, and purposeful talk (Alexander, 2018; Howe et al., 2019). Collective and reciprocal learning occur as knowledge is co-constructed through question-and-answer exchanges. Supportive and deliberative elements from the chatbot, emerge through empathetic language and explanations connecting principles to real-life examples, while cumulative and purposeful talk links new knowledge to prior learning and keeps discussions focused on historical understanding and ethical reasoning.

4. PLAY TESTING

Prior to the playtest, students were provided with reading materials a week in advance to build background knowledge of the historical figures, helping them pose more meaningful questions. At the start of each session, they had five minutes to review the day's material. A total of 52 middle-school students (21 from Grade 6, 18 from Grade 7, and 13 from Grade 8) from an urban English-medium school participated voluntarily with parental consent.

The playtesting aimed to explore how students engage with personified chatbot prototypes representing historical figures, including Mahatma Gandhi, Benjamin Franklin, Leonardo da Vinci, Chanakya, and Albert Einstein. Over five classroom sessions conducted during regular school hours, each day focused on a different persona, supervised by both the researcher and the classroom teacher. Students used individual devices to ensure authentic one-to-one interactions with the chatbots, which were exploratory in nature, allowing them to freely ask questions, probe ideas, and take notes on insights of interest. Each student interacted with all five chatbots across non-consecutive days to avoid cognitive overload while maintaining balanced exposure. All conversations were automatically logged by the system, capturing both student questions and chatbot responses. These logs served as the primary dataset for analysis, offering insights into the types of questions posed, the quality of chatbot responses, and the potential of the system as an engaging educational tool for learning history.

5. FINDING AND DISCUSSION

This study followed a case study approach (Tisdell E. et. al. 2025) and focused on a single student who participated over five days and interacted with each of the persona-based historical chatbots. Following the content analysis (Bengtsson, Mariette., 2016) of the student-chatbot interactions, this case demonstrates both the nature of the student's questions and how the chatbots replied, illustrating their conversational strategies and guidance.

On the first day, the student started the conversation with the Gandhi chatbot by asking about historical events and personal motivations, such as "*Why did you choose to fight for freedom?*" and "*What roles did your parents play in shaping your values?*". The bot responded with a mix of historical context, moral reasoning, and prompts that encouraged the student to reflect personally, such as "*What do you think about that?*" or "*What values do you cherish from your own family?*". Students asked hypothetical questions, such as "*What if you didn't face challenges in Africa?*" This invited counterfactual reasoning, enabling the student to connect historical events with choices and character, demonstrating both cognitive and emotional engagement.

On the second day, the student engaged with the Einstein chatbot. The student asked about the inspiration behind the theory of relativity, other scientific contributions, collaborative influences, and personal challenges like speech difficulties, for example, *“What inspired you to invent the theory of relativity?”*, *“Did someone help you for this?”*, *“You had speech problems, did you face any challenges?”*, *“What are your other inventions?”* was complemented by reflective and hypothetical inquiries like *“Could this theory of relativity be used for negative purposes?”* and *“Could you say any words for the younger generation?”* The bot provided historically and scientifically grounded responses while prompting the student to consider broader social, ethical, and personal implications of scientific work, for instance, *“How do you think you would respond if faced with challenges or opportunities?”*, *“..Keep exploring, and who knows what ideas you might come up with next!”*. Likewise, after every question from the student, the bot gives the motivations, value to the student so that the student keeps asking questions.

The third-day conversation with the Benjamin Franklin chatbot also showed a mix of question types. The student began with factual queries about Franklin’s role in the Revolutionary War, inventions, and writings (*“Which books did you write?”*, *“What inventions did you create?”*) before shifting to ethical and reflective ones (*“What were your views on slavery?”*, *“What advice would you give to the younger generation?”*). The chatbot combined historical accuracy, anecdotes, and humor, recounting how he left school at ten to work in his family’s candle-making business while stressing perseverance. It emphasized civic responsibility and collaboration through quotes like *“An ounce of prevention is worth a pound of cure... When we share knowledge, we uplift each other.”* Through such storytelling and practical examples, the bot made abstract concepts like responsibility, collaboration, and self-education accessible, encouraging both intellectual and moral reasoning.

On the fourth day, the interaction with the Leonardo da Vinci chatbot included both factual and reflective exchanges. The student asked questions such as *“How did you become an artist?”*, *“What inspired you to draw Mona Lisa?”*, and *“Who supported you with all your artworks?”*, to which the chatbot responded with historical details about his apprenticeship, patrons, and inspirations. As the conversation progressed, the student posed deeper inquiries like *“Why didn’t you create working models of your inventions?”*, *“Does art require mathematics?”*, and *“How do you investigate a subject?”*, and the chatbot replied with explanations that combined historical context, artistic philosophy, and scientific reasoning, such as *“I observed, sketched, experimented, and reflected to investigate each subject thoroughly.”* When asked *“Can you give unique mediums to use for art?”*. The chatbot offered motivational guidance and practical suggestions, closing with encouragement like *“Remember, experimenting with different mediums is part of the fun! Don’t be afraid to try new things!”* Throughout the conversation, the bot consistently tried to encourage the student to think further by closing its replies with advice, reflection, or a moral takeaway. The responses such as, *“I painted ‘The Last Supper’ between 1495 and 1498. It depicts the moment Jesus announces that one of his disciples will betray him”*, *“When investigating a subject, I like to immerse myself in the details! ... Observation, Research, Sketching, Experimentation, Reflection.”* and *“Remember, every artist’s path is unique, and curiosity is your greatest tool!”*, illustrate the blend of historical fact-sharing, methodological explanation, and motivational prompting, showing an intentional design to both inform and inspire.

Finally, on the fifth day, the Chanakya chatbot encouraged historical, strategic, ethical, and practical inquiry. The student asked foundational questions (*“Why are you famous?”*, *“How can a ruler ensure equality and inclusivity?”*, *“What are possible consequences when an economy falls?”*) and higher-order ones (*“How to score in every subject?”*). The chatbot gave structured, actionable responses combining historical examples and practical guidance, prompting strategic thinking, ethical evaluation, and reflection. Leadership weaknesses were listed as *“inconsistency, lack of confidence, poor communication... and favoritism”*, with advice like *“By organizing study, actively learning, and adapting, you can improve your results...”*. Historical examples illustrated governance (*“I helped Chandragupta Maurya establish the Maurya Empire... teaching him about ruling wisely...”*), ethical reasoning (*“A ruler can maintain ethics by... promoting openness to reduce corruption...”*), and strategic thinking (*“Being strategic in leadership is very important... success hinges on thinking ahead...”*). These responses not only informed the student about ancient statecraft but also encouraged reflection on fairness, foresight, and personal growth in modern contexts.

Overall, the interactions showed that students engaged most when the chatbots provided contextually rich, persona-driven responses paired with reflective prompts. Across all five days, the chatbots consistently combined historical, factual, and contextual information with guidance, motivational cues, storytelling, and practical suggestions drawn from the personas’ own life experiences. Each bot adapted its responses to its persona and topic, blending information with prompts that encouraged reflection, exploration, and continued questioning.

By emulating historical figures, such chatbots provide students with a unique opportunity to interact with the past in a dynamic and personalized manner. For example, educators have employed AI chatbots modeled on figures like Abraham Lincoln to facilitate discussions on complex historical events, enabling students to pose questions such as, “*Why did you hesitate before issuing the Emancipation Proclamation?*” (SchoolAI, 2025). Similarly, Park et al. (2025) highlight that integrating character-specific responses and personalization features into chatbots can significantly enhance engagement and curiosity.

With this preliminary study, we focused was on documenting designing of the persona based chatbot, and how students interacted with the chatbots; however, future research will examine in greater depth the cognitive levels of student questions, the engagement trajectory over time, and the role of learner mindset in shaping the depth of inquiry.

6. CONCLUSION

The study documents the design and implementation of persona-based historical chatbots and the ways in which students interacted with them over five classroom sessions. Students engaged with chatbots representing Mahatma Gandhi, Albert Einstein, Benjamin Franklin, Leonardo da Vinci, and Chanakya. Interaction logs showed that students began with factual questions and gradually moved toward reflective, ethical, and hypothetical inquiries. The persona-based chatbots provided contextually rich responses aligned with each historical figure’s communication style. Students asked mixed-level questions spanning factual understanding, ethical reasoning, interdisciplinary connections, and personal reflection. The chatbot responded with contextually rich narratives, ethical reasoning, and prompts that encouraged personal reflection, supporting iterative dialogue throughout the interaction. Such interactions can support the student to progress from basic knowledge acquisition to analytical thinking and moral evaluation, integrating factual understanding with ethical and personal application. It should be noted that the insights from the student- chatbot interactions are based on a single case study, which calls for further investigations by extending the analyses to the rest of student cases.

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